Course Project for SE

Front-End



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Version: 1.0

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PROJECT DELIVERABLE D3

System Requirements Specification

Ludo Game Platform (Front End)

Course	Software Engineering (SE), Fall 2026
Team	Power Puff Girls
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Prepared as the system-requirements deliverable for the front-end component. This specification is based on the team's D2 requirements elicitation document; unresolved client decisions remain explicitly marked as items to confirm.

Document control

Field	Value
Document	Project Deliverable D3 — System Requirements Specification
Component	Ludo Game Platform (Front End)
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1. Scope statement

This document specifies the system requirements for the front end of the Ludo Game Platform (LGP), developed by team Power Puff Girls for the Software Engineering course, Fall 2026. It translates the user needs recorded in Deliverable D2 into a more precise description of the front-end component's responsibilities, interfaces, functional behaviour and quality constraints. The document is intended to help other project teams understand the component and integrate with it during the next phase of the project.

The front end provides the player-facing experience for a digital, multiplayer, race-based board game inspired by traditional Ludo. It supports game setup for 2, 3 or 4 players; Human vs Human and Human vs Computer play; player names and colours; board, pieces and dice presentation; turn and progress information; move selection and feedback; winner presentation; and game controls and display settings. The interface is expected to request rule decisions and move validation from the game-logic component and computer moves from the game-AI component. Those components are outside the scope of this front-end specification.

The front end is distinguished by an uncluttered, single-screen game view, clear indication of the active player, immediate and understandable feedback for invalid actions, visual guidance for legal moves, accessible presentation of player progress, and configurable sound, animation and theme settings. The exact rules for special dice rolls, captures, safe squares, stacking/blocking and entry into the final destination remain subject to client confirmation, as recorded in D2. This document does not claim that those unresolved rules have already been implemented.

1.1 Scope boundaries

Included: player-facing screens and controls; setup and mode selection; board and piece rendering; dice interaction and result display; turn/status/message presentation; input handling; settings; presentation of game outcomes; and the front-end interfaces needed to communicate with game logic and game AI.

Excluded: authoritative rule computation, legal-move validation, dice randomness generation, AI strategy, persistent online accounts, matchmaking, remote multiplayer networking, server hosting and database implementation. These are not specified as front-end responsibilities in D2. If required by the wider project, they must be agreed and specified by the relevant component teams.

1.2 Intended audience

The primary audience is the development team responsible for the front end and the teams responsible for game logic and game AI. The instructor, teaching assistant and client representatives may use this document to review scope, clarify open decisions and assess the component during integration.

1.3 Contact information

Group: Power Puff Girls. Members: Zainab Tariq (BSDSF24M004), Saneela Aziz (BSDSF24M022) and Hifza Zia (BSDSF24M034). D2 does not provide email addresses or telephone numbers; therefore, no contact details have been invented. The team should add an agreed group email or other approved contact channel before public release.

2. Definitions and game rules

The following terms are used consistently throughout this specification. Where D2 leaves a rule undecided, the definition describes the interface concept without selecting an unapproved rule.

Ludo Game Platform (LGP)	The overall digital board-game product. This document specifies only its front-end component.
Front end	The player-facing part of the platform: screens, board, pieces, dice control, status display, messages and settings.
Player	A human participant or a computer-controlled participant occupying a seat in a game.
Human vs Human (HvH)	A game mode in which the participating seats are controlled by people.
Human vs Computer (HvC)	A game mode in which at least one seat is controlled by the computer player. D2 also proposes mixed human/computer seats, subject to confirmation.
Seat	A position in the game setup assigned to a player, with a name, colour and controller type where supported.
Piece / token	A movable game object belonging to one player. D2 refers to pieces in a starting area, on the track and at the destination.
Starting area / home	The area in which a player's pieces wait before entering the track.
Track	The sequence of board spaces along which pieces move.
Destination / finish	The final area or goal that a player's pieces must reach to complete the game.
Dice roll	The result used to determine the number of spaces a selected piece may move. The front end displays the result; the unbiased generation is an external game-logic responsibility.
Active player / turn	The player currently permitted to perform the next game action.
Valid move	A move accepted by the game-logic component under the confirmed game rules.
Invalid move	A requested move rejected by game logic. The front end presents a clear explanation when one is available.
Capture	A rule-governed event in which a piece landing on an opponent's piece sends that piece to its starting area; D2 marks this rule as to confirm.
Safe square	A board space on which capture may be restricted; the exact set and effect require confirmation.
Stacking / blocking	The rule governing two or more pieces occupying the same square. D2 leaves the behaviour to confirm.
Computer player / AI	The component that chooses a move for a computer-controlled seat. It is separate from the front end.
Game state	The current authoritative information about players, pieces, turn, dice result, legal moves, scores and game status.
Session score	A summary of results across consecutive rematches, as requested in D2.

2.1 Game flow represented by the front end

At a high level, a player starts a game, selects the number of seats and mode, enters player names and colours, and begins the match. During play, the interface shows whose turn it is, accepts a dice-roll action, displays the result, highlights the pieces reported as movable, and submits the selected move for validation. It then updates the visible board and status from the authoritative game-state response. When a player completes the game, the interface presents the winner and any available ranking or session score, and offers restart or new-game controls.

2.2 Rules requiring client confirmation

Topic	Current D2 position	Required decision before final integration
First player	FR-08: selection method to confirm	Highest initial roll, fixed order, or another agreed rule.
Special roll	FR-12: roll and effect to confirm	Which result triggers an extra turn and/or releases a piece.
No legal move	FR-13	Confirm whether the turn is skipped automatically and whether a message is shown.
Finish entry	FR-20: exact-roll rule to confirm	Whether a piece needs an exact roll to enter the destination.
Capture	FR-21: to confirm	Whether landing on an opponent always captures and how return to home works.
Safe squares	FR-22: to confirm	Identify safe squares and define their capture protection.
Multiple pieces per square	FR-24: to confirm	Choose stacking, blocking, or another rule and define its effect.
Final ranking	FR-28: to confirm	Define ranking/tie handling after the winner is decided.
Mixed seats / difficulty	FR-05 and FR-31: to confirm	Confirm mixed human/computer seats and available AI difficulty levels.

3. Components and system architecture

The diagrams describe the logical architecture of the Ludo front end and its collaboration with the game-logic and game-AI components. They show conceptual responsibilities and interfaces; they do not imply a particular deployment topology.

3.1 Simplified component view

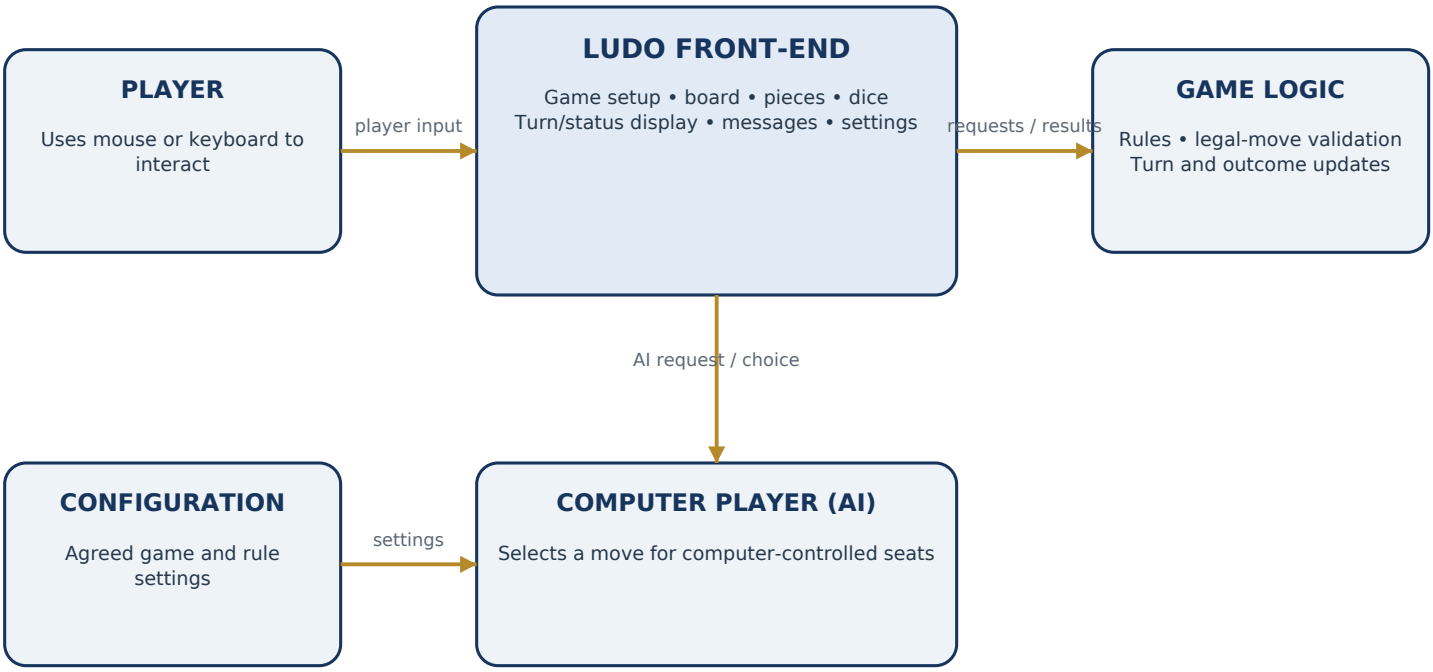


Figure 1. Simplified logical context of the Ludo front end.

How the components work together. The player interacts with the front end. The front end sends roll and move requests to game logic and displays the authoritative results returned by that component. When a computer-controlled seat is active, the front end requests a proposed move from the AI; game logic validates the selected move. Configuration supplies agreed game settings. The front end presents outcomes but does not independently change or override game rules.

Diagram key

Direction of information / request flow

Logical view only — detailed interface contracts are defined during integration.

3.2 Front-end internal architecture

The front end is divided into presentation, interaction, state-management and integration responsibilities. The game-logic and game-AI components remain external dependencies.

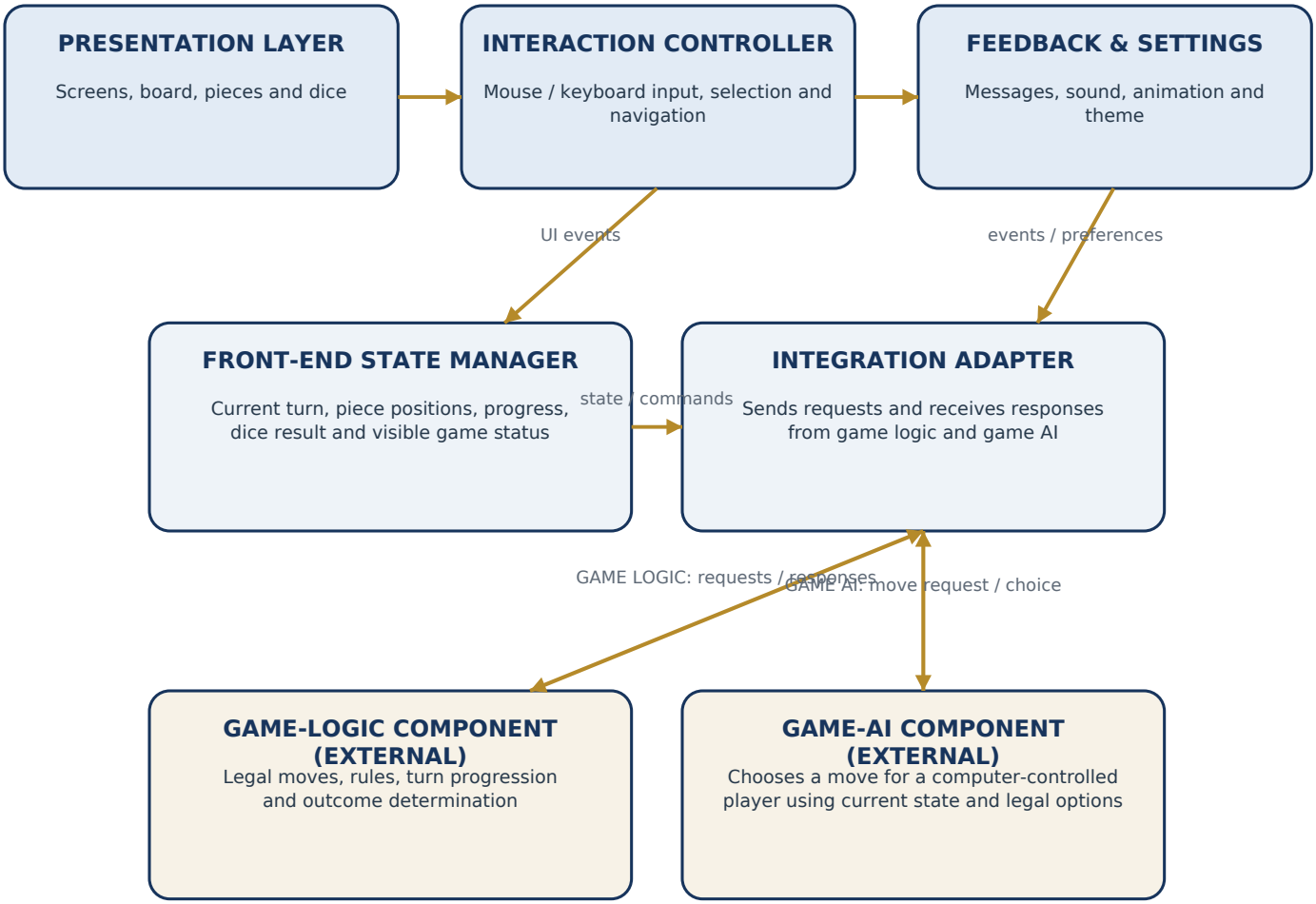


Figure 2. Proposed internal decomposition and external interfaces.

Component	Main responsibility
Presentation layer	Renders the menu, setup, board, pieces, dice, player panel and winner screen.
Interaction controller	Handles clicks, taps and keyboard actions; manages selection and navigation.
Front-end state manager	Maintains the visible snapshot of players, turn, pieces, progress, messages and game status.
Integration adapter	Isolates the front end from external component details; sends requests and receives results/errors.
Feedback & settings	Shows invalid-move explanations and action messages; applies sound, animation and theme preferences.
Game logic (external)	Owns authoritative rules, legal moves, turn progression, state transitions and win detection.
Game AI (external)	Selects a proposed move for a computer-controlled player; game logic validates it.

3.3 Interface contracts

Front end ↔ game logic: the front end submits setup choices, roll requests and selected-piece move requests. Game logic returns the current game state, legal-move options, accepted/rejected status, explanations where available, turn changes, capture events and completion status. The game-logic component is the source of truth for rule enforcement.

Front end ↔ game AI: when a computer-controlled seat becomes active, the front end (or the agreed orchestration layer) provides the AI with the current state and available legal moves. The AI returns a move choice. The chosen move is still submitted to game logic for authoritative validation; the front end displays the resulting accepted state.

Configuration: the values for number of pieces, board layout, safe squares and special-roll behaviour should be supplied by an agreed configuration/rules interface rather than embedded as assumptions in the presentation code. D2 requests separation of rules/settings from interface code (NFR-12).

3.4 State and error handling

The interface should render from the latest accepted game-state response. A rejected move must not cause the displayed authoritative state to advance. If an external component returns an error or an unexpected response, the front end should preserve the last known valid state, show a concise message and allow the player to retry or return to a safe screen. This approach supports D2's reliability requirement that invalid moves and unexpected input do not lose the current game state (NFR-04).

4. Functional requirements

The requirements below specify the behaviour expected from the front-end component. They are derived from D2 user stories and focus on system behaviour rather than describing only the user's actions. The game logic remains authoritative for rules and validation. The creator/date fields are included for traceability; the requirements are prepared by the project team for this D3 version.

4.1 Requirement format and traceability

Each requirement has a unique identifier, name, source D2 user requirement, primary actor, description, trigger, preconditions, postconditions, normal flow, alternative/exception handling and acceptance checks. Items marked TO CONFIRM depend on decisions not yet supplied in D2.

FR-SYS-01 — Start and configure a new game

Field	Specification
Created by / date	Power Puff Girls / 2026-11-04
Primary actor / source	Player / FR-01, FR-02, FR-03, FR-04, FR-06
Description	The front end shall present a new-game setup flow in which the player can select 2, 3 or 4 seats, choose Human vs Human or Human vs Computer mode, and enter a display name and colour for each configured player.
Trigger	Player selects New Game.
Preconditions	The main screen is visible; the game is not in an active match.
Normal flow	1. Show the supported player-count choices. 2. Show the supported game modes. 3. Create the required seat entries. 4. Collect a name and colour for each seat. 5. Submit the setup configuration to game logic. 6. On success, show the initial board and player status.
Alternative flows / exceptions	Reject missing or duplicate values only if such validation is agreed; otherwise display the validation response returned by the setup service. If setup fails, retain the entered configuration and show an actionable error.
Postconditions / acceptance	Each supported player count can be selected; the setup request contains the selected mode and all seat names/colours; a successful response opens the game view.

FR-SYS-02 — Present the active turn and dice result

Field	Specification
Created by / date	Power Puff Girls / 2026-11-04
Primary actor / source	Player / FR-08, FR-09, FR-10, FR-11
Description	The front end shall display the active player's identity and provide a clear dice control. After a roll request, it shall display the result returned by game logic and update the active-player indicator when the returned state changes.
Trigger	An active game is ready for a player's turn.
Preconditions	A valid game state exists and the current turn permits a roll.
Normal flow	1. Highlight the active player. 2. Enable the roll control when permitted. 3. On activation, send one roll request. 4. Display the returned dice result. 5. Refresh turn and action status from the returned state.
Alternative flows / exceptions	Prevent repeated roll submissions while a request is pending. If a roll request fails, keep the previous state and show a retry message. First-player selection method and special-roll effects are TO CONFIRM.
Postconditions / acceptance	The active player is always identifiable; one user action produces at most one pending roll request; the displayed result matches the response.

FR-SYS-03 — Show legal piece choices and submit a move

Field	Specification
Created by / date	Power Puff Girls / 2026-11-04
Primary actor / source	Player / FR-14, FR-15, FR-16, FR-18
Description	After receiving a dice result and legal-move data, the front end shall visually distinguish pieces that game logic reports as movable. It shall allow the player to select one of those pieces, preview the reported destination when supported, and submit the selected move for authoritative validation.
Trigger	Game logic returns a dice result and legal-move options.
Preconditions	The game is active; the roll request has completed; legal-move data is available.
Normal flow	1. Receive the legal-move list. 2. Highlight each eligible piece. 3. Let the player select an eligible piece. 4. Show the destination preview if supplied. 5. Submit the selected piece identifier and move request. 6. Wait for the authoritative response before updating the board.
Alternative flows / exceptions	If no legal move is returned, follow the agreed no-move behaviour (TO CONFIRM). If a selected piece is no longer legal, show the returned explanation and refresh the legal options. Do not move a piece optimistically before acceptance.
Postconditions / acceptance	Only pieces in the returned legal list are presented as selectable; a rejected move does not change the authoritative board state.

FR-SYS-04 — Explain invalid moves and preserve state

Field	Specification
Created by / date	Power Puff Girls / 2026-11-04
Primary actor / source	Player / FR-17, NFR-04
Description	When game logic rejects a move or the interface receives invalid input, the front end shall display a concise, understandable explanation when available and preserve the last accepted game state.
Trigger	A move is rejected or an invalid action is received.
Preconditions	A game is active and a last accepted state is available.
Normal flow	1. Receive a rejection or detect malformed UI input. 2. Keep the last accepted state unchanged. 3. Display the explanation returned by game logic, or a generic non-technical message if none is supplied. 4. Re-enable appropriate controls when safe.
Alternative flows / exceptions	If the response is incomplete or unavailable, do not infer a move outcome; show a temporary error and retain the last known state. Provide a retry path where appropriate.
Postconditions / acceptance	An invalid action never silently changes piece positions, turn ownership or scores; the user receives visible feedback.

FR-SYS-05 — Animate accepted movement and show game events

Field	Specification
Created by / date	Power Puff Girls / 2026-11-04
Primary actor / source	Player / FR-19, FR-23, FR-36, FR-38
Description	After an accepted move, the front end shall update the board to the returned state, present movement in a step-by-step animation when enabled, and show a short event message for relevant actions such as a capture when that event is reported.
Trigger	Game logic confirms a move and returns updated state/events.
Preconditions	A move has been accepted and the response contains the updated state.
Normal flow	1. Read the accepted state and event list. 2. If animation is enabled, animate the piece along the returned path. 3. Otherwise render the final position immediately. 4. Display a concise description of the last action. 5. Refresh player progress and turn information.
Alternative flows / exceptions	If the event list is absent, do not invent an event. Capture display and the underlying capture rule are TO CONFIRM in D2.
Postconditions / acceptance	The final rendered board matches the accepted state; disabling animation does not change the result; reported events are visible in the message area.

FR-SYS-06 — Display player progress and game completion

Field	Specification
Created by / date	Power Puff Girls / 2026-11-04
Primary actor / source	Player / FR-25, FR-26, FR-27, FR-28, FR-29
Description	The front end shall display each player's piece counts in the starting area, on the track and at the destination, and shall present a winner screen when game logic reports that a player has completed the game.
Trigger	Game state changes or game logic reports completion.
Preconditions	A game is active or has reached a completed state.
Normal flow	1. Read the progress data for each player. 2. Display the three requested piece counts. 3. On completion, display the winner and final game status. 4. Display final ranking if supplied and confirmed. 5. Maintain or display a session score across rematches if that feature is enabled.
Alternative flows / exceptions	If ranking/tie rules are not confirmed, show only the winner information returned by game logic. Do not calculate an independent winner in the front end.
Postconditions / acceptance	Displayed counts match the returned state; winner screen appears only on a completion response; optional ranking/session score is not fabricated.

FR-SYS-07 — Request and present computer-player moves

Field	Specification
Created by / date	Power Puff Girls / 2026-11-04
Primary actor / source	Player / Computer player / FR-30, FR-31
Description	When a computer-controlled seat becomes active, the front end shall request a move from the game-AI component using the current state and available legal options, then submit the returned choice to game logic and display the accepted result.
Trigger	A computer-controlled player's turn begins.
Preconditions	The seat is configured as computer-controlled; the AI interface is available; legal options/current state are available.
Normal flow	1. Detect that the active seat is computer-controlled. 2. Provide the current state and legal options to AI. 3. Receive the AI-selected move. 4. Submit the move to game logic. 5. Render the accepted state and event feedback.
Alternative flows / exceptions	If AI does not respond within the agreed timeout or returns an invalid choice, show a recoverable error and request a retry or use the fallback behaviour agreed by the project. Difficulty selection is TO CONFIRM.
Postconditions / acceptance	The computer move is validated by game logic before it is displayed as completed; AI failure does not corrupt the game state.

FR-SYS-08 — Control game restart, exit and rematch

Field	Specification
Created by / date	Power Puff Girls / 2026-11-04
Primary actor / source	Player / FR-32, FR-33, FR-34
Description	The front end shall provide controls to restart the current game, leave the current game and start a new game after completion. It shall request confirmation before an action that would discard active-game progress.
Trigger	Player selects Restart, Exit or New Game.
Preconditions	The relevant game screen is visible.
Normal flow	1. Receive the selected control action. 2. If active progress may be lost, show a confirmation prompt. 3. If cancelled, return to the current game unchanged. 4. If confirmed, send the appropriate restart/new-game request or return to the main screen. 5. Render the resulting state.
Alternative flows / exceptions	If the restart request fails, retain the current state and show an error. A rematch may update the session score if that feature is enabled and confirmed.
Postconditions / acceptance	Cancel leaves the game unchanged; confirm performs only the selected action; a failed request does not erase the current state.

FR-SYS-09 — Provide board, player and action information in one view

Field	Specification
Created by / date	Power Puff Girls / 2026-11-04
Primary actor / source	Player / FR-35, FR-36
Description	During gameplay, the front end shall present the board, starting areas, track, destination, pieces, dice control, active-player indicator, player progress and a concise game-information/message area in a coherent game screen.
Trigger	Player opens or enters the active game view.
Preconditions	A game state has been initialized.
Normal flow	1. Render the board and all player pieces. 2. Display the dice control and latest result. 3. Display active player and turn status. 4. Display progress counts for each player. 5. Display the latest action message.
Alternative flows / exceptions	If the available window is too small to show all elements at once, adapt the layout without hiding essential game status; exact mobile support is TO CONFIRM.
Postconditions / acceptance	All required game information is available without navigating away from the active game screen at the supported desktop sizes.

FR-SYS-10 — Manage sound, animation and visual theme settings

Field	Specification
Created by / date	Power Puff Girls / 2026-11-04
Primary actor / source	Player / FR-37, FR-38, FR-39
Description	The front end shall allow players to turn sound effects on or off, skip or disable movement animations, and switch between light and dark visual themes.
Trigger	Player opens the settings control or changes a quick setting.
Preconditions	The front end is running.
Normal flow	1. Show the three settings. 2. Let the player change each setting. 3. Apply the change to subsequent relevant interface events. 4. Keep the current game state unchanged.
Alternative flows / exceptions	If a setting cannot be applied, retain the previous value and show a clear message. Persistence of preferences across sessions is not specified in D2 and requires confirmation.
Postconditions / acceptance	Sound can be muted; animations can be skipped/disabled without changing game outcomes; theme can be switched without losing game state.

5. Non-functional requirements

Non-functional requirements define quality constraints and operating expectations for the front-end component. They are aligned with D2 NFR-01 to NFR-14. Values explicitly described as provisional in D2 remain subject to confirmation; no unverified performance or platform result is claimed.

NFR-SYS-01 — Interaction responsiveness

Attribute	Specification
Category	Performance
Requirement	The front end shall provide visible acknowledgement of a user action and complete its own response handling within 1 second under the agreed normal test environment. The 1-second target is provisional, as noted in D2 NFR-01; the time boundary for external game-logic processing must be agreed.
Verification	Measure from user action to visible acknowledgement/response using repeatable tests. Confirm whether the 1-second target includes external service time.
D2 traceability	NFR-01

NFR-SYS-02 — Computer move response

Attribute	Specification
Category	Performance
Requirement	The computer player shall return a move within 2 seconds under the agreed normal test environment. The AI response target is inherited from D2 NFR-02 and is subject to confirmation.
Verification	Record request-to-response times across representative turns; report the percentile/maximum agreed by the team.
D2 traceability	NFR-02

NFR-SYS-03 — Stable normal gameplay

Attribute	Specification
Category	Reliability
Requirement	The front end shall not crash during normal gameplay and shall handle routine screen transitions, rolls, selections and accepted moves without terminating the session.
Verification	Run repeated end-to-end gameplay sessions and verify that the application remains usable.
D2 traceability	NFR-03

NFR-SYS-04 — Preserve state after invalid input

Attribute	Specification
Category	Reliability
Requirement	The front end shall handle invalid moves and unexpected user input without losing the last accepted game state.
Verification	Test malformed, repeated and rejected actions; compare the state before and after rejection.
D2 traceability	NFR-04

NFR-SYS-05 — Fair dice presentation and source of randomness

Attribute	Specification
Category	Fairness
Requirement	The front end shall display the dice outcome supplied by the authoritative game-logic component and shall not bias, alter or substitute the returned result. The unbiased random-number generation itself belongs to game logic, consistent with the D2 boundary.
Verification	Verify that the displayed value exactly matches the returned value; fairness of the generator must be tested by the game-logic team.
D2 traceability	NFR-05

NFR-SYS-06 — Understandable first-turn experience

Attribute	Specification
Category	Usability
Requirement	A first-time player shall be able to understand the short rules summary and complete a first turn without outside help, subject to the usability measure being agreed.
Verification	Conduct a first-use test with representative first-time players; record completion and assistance required. Measurement threshold is TO CONFIRM.
D2 traceability	NFR-06

NFR-SYS-07 — Readable and distinguishable game elements

Attribute	Specification
Category	Usability
Requirement	Text, pieces and board spaces shall remain readable on supported screens. Player identity shall not rely on colour alone; use a distinguishable label, shape or other visual cue as well.
Verification	Inspect supported desktop window sizes and test distinguishability in grayscale or with colour-vision simulation.
D2 traceability	NFR-07

NFR-SYS-08 — Simple language and keyboard operation

Attribute	Specification
Category	Usability
Requirement	Rules and messages shall use short, simple wording. Core game actions shall be operable with a keyboard as well as a mouse, as required by D2 NFR-08 and NFR-09.
Verification	Review interface text for clarity; complete setup, roll, move selection and game controls using keyboard-only navigation.
D2 traceability	NFR-08, NFR-09

NFR-SYS-09 — Desktop browser compatibility

Attribute	Specification
Category	Platform
Requirement	The front end shall run in the latest agreed versions of Chrome, Firefox and Edge on desktop without requiring installation. The exact browser versions and test environment are TO CONFIRM.
Verification	Execute the agreed smoke and gameplay test suite in each supported browser.
D2 traceability	NFR-10

NFR-SYS-10 — Responsive layout

Attribute	Specification
Category	Platform
Requirement	The interface shall adapt to different desktop window sizes while keeping essential controls and game information accessible. Whether mobile browser support is required remains TO CONFIRM.
Verification	Resize the browser window through the agreed supported range and verify that no essential control is obscured.
D2 traceability	NFR-11

NFR-SYS-11 — Separate rules/configuration from presentation

Attribute	Specification
Category	Extensibility
Requirement	Game rules and settings, including number of pieces, board size, safe squares and special rolls, shall be represented through configuration or a defined rules interface rather than being hard-coded into presentation components.
Verification	Change a configuration value in a controlled test and verify that the UI reflects the supplied configuration without redesigning the presentation layer.
D2 traceability	NFR-12

NFR-SYS-12 — Defined replaceable component interfaces

Attribute	Specification
Category	Extensibility
Requirement	The front end shall communicate with game logic and the computer player through defined interfaces so that either external component can be replaced or extended without requiring a redesign of the front-end presentation layer.
Verification	Use interface-level tests or mocks to verify that the front end depends on documented requests and responses rather than internal implementation details.
D2 traceability	NFR-13

NFR-SYS-13 — Maintainable shared codebase

Attribute	Specification
Category	Maintainability
Requirement	The front-end code shall be structured and commented sufficiently for all three team members to understand, review and modify it.
Verification	Conduct peer review; verify consistent module boundaries, naming and documentation for key components.
D2 traceability	NFR-14

6. Assumptions, dependencies and open decisions

This section separates what D2 explicitly requires from what must be agreed before implementation or integration. It prevents the front end from silently taking ownership of game rules that D2 assigns to other components or leaves undecided.

6.1 Assumptions used in this specification

• Game logic is the authoritative source for dice outcomes, legal moves, turn changes, piece positions, captures and win detection.

• The front end receives sufficient state and event information to render the board, player progress and action messages.

• The computer-player component returns a proposed move; game logic validates that move before it becomes part of the accepted game state.

• The initial supported player counts are 2, 3 and 4, as stated in D2.

• The required baseline modes are Human vs Human and Human vs Computer. Mixed human/computer seats are not treated as confirmed functionality.

6.2 Dependencies

Dependency	Required agreement / information
Game logic	Request/response schema for setup, roll, legal moves, move submission, rejection reasons, updated state, events and completion.
Game AI	Input state format, legal-move representation, move response format, timeout behaviour and optional difficulty parameter.
Board/rules definition	Board coordinates, track order, home/destination layout, piece count, player colour mapping and configurable rule values.
Integration environment	How the front end invokes the other components (direct module/API contract), error format and expected response timing.
Client / teaching team	Confirmation of all open rules listed in Section 2.2 and acceptance criteria for provisional performance/usability targets.

6.3 Open decisions checklist

☐ Confirm first-player selection and special-roll effects.

☐ Confirm no-valid-move behaviour and whether an automatic skip message is required.

☐ Confirm exact-roll requirement for entering the final destination.

☐ Confirm capture rules, safe squares and stacking/blocking behaviour.

☐ Confirm final ranking, tie handling and session-score rules.

☐ Confirm mixed human/computer seats and AI difficulty options.

☐ Confirm whether settings persist across sessions.

☐ Confirm supported desktop window range and whether mobile support is required.

☐ Confirm exact browser versions and performance measurement boundaries.

☐ Add an approved group contact email or contact channel before the document is released to the class.

6.4 Limitations

D2 is a requirements-elicitation deliverable and does not provide implementation details, source code, API schemas, screenshots of the completed front end or test results. Accordingly, this D3 specifies expected behaviour and a logical integration architecture; it does not assert that the listed features have been implemented or verified. Any implementation-specific details should be added only after the team confirms them.

A. Changelog

This is the initial D3 specification based on the team's D2 baseline. Since D2, the requirements have been elaborated into front-end system requirements, a logical component architecture, interface responsibilities, verification criteria and an explicit list of unresolved decisions.

Version	Date	Change
v1.0	2026-11-04	Initial D3 specification. Derived from D2 v1 dated 2026-10-18; added logical architecture, definitions, detailed functional requirements, non-functional requirements, traceability and open-decision register.

B. Grading — Individual contributions

The D3 brief requires each group member to be responsible for at least two functional requirements and three non-functional requirements. The following allocation records responsibility for drafting the requirements in this document. It should be reviewed and confirmed by the team before submission; the allocation is not independently verified by the source documents.

Team member	Functional requirements drafted	Non-functional requirements drafted
Zainab Tariq BSDSF24M004	FR-SYS-01, FR-SYS-02, FR-SYS-03	NFR-SYS-01, NFR-SYS-02, NFR-SYS-03, NFR-SYS-04
Saneela Aziz BSDSF24M022	FR-SYS-04, FR-SYS-05, FR-SYS-06	NFR-SYS-05, NFR-SYS-06, NFR-SYS-07, NFR-SYS-08
Hifza Zia BSDSF24M034	FR-SYS-07, FR-SYS-08, FR-SYS-09, FR-SYS-10	NFR-SYS-09, NFR-SYS-10, NFR-SYS-11, NFR-SYS-12, NFR-SYS-13

C. D2 traceability summary

The following mapping shows how the major D2 requirement groups are addressed in D3. It is a navigation aid; the detailed source identifiers are included in each system requirement.

D2 user requirements	D3 coverage
FR-01 to FR-07 — Setup and modes	FR-SYS-01; definitions; open decisions
FR-08 to FR-13 — Turns and dice	FR-SYS-02; Section 2.2 rules checklist
FR-14 to FR-20 — Movement and valid moves	FR-SYS-03, FR-SYS-04; Section 2.2
FR-21 to FR-24 — Player interaction/capture	FR-SYS-05; Section 2.2
FR-25 to FR-29 — Progress and winning	FR-SYS-06
FR-30 to FR-31 — Computer player	FR-SYS-07
FR-32 to FR-34 — Game control	FR-SYS-08
FR-35 to FR-39 — Interface and settings	FR-SYS-09, FR-SYS-10
NFR-01 to NFR-14	NFR-SYS-01 to NFR-SYS-13

End of Deliverable D3